PARKINSON'S DISEASE DETECTION FROM GAIT FORCE SIGNALS WITH A STATE-SPACE (MAMBA) MODEL: RESULTS AND THE ROLE OF THE EVALUATION PROTOCOL

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ABSTRACT

Published accuracies for Parkinson's disease detection from gait force recordings span 70–99 % on the PhysioNet database, a spread wider than the differences between the models involved. We classify patients against controls with a Mamba state-space model on 100 subjects, under two evaluation protocols on identical data. Held to subject-disjoint splits, with the configuration chosen on validation rather than on test, the model reaches 75 ± 4 % accuracy and 0.85 AUC on both feet, level with a tuned Random Forest and ahead of SVM-RBF, 1D-ResNet and LSTM baselines. The gain comes from describing each second by its channel statistics instead of feeding raw samples, which also removes a training collapse seen on two folds. Re-running the same baselines with window-wise splits, where windows of one subject fall on both sides, lifts Random Forest accuracy from 73 % to 98 %, a number we report only as a caution.

***Index Terms*—** Parkinson's disease, gait analysis, state-space models, Mamba, data leakage

1. INTRODUCTION

Parkinson's disease (PD) is the second most common neurodegenerative disorder, and disturbed gait is one of its earliest and most visible motor signs. Clinical scales such as the UPDRS rely on expert observation and are coarsely quantized, which is why sensor-based measures are attractive. Instrumented insoles record the vertical ground reaction force (VGRF) under each foot cheaply and at high rate, and the resulting series carries the stride-timing irregularity and left-right asymmetry that characterize parkinsonian gait [1, 2].

Accuracies published on the PhysioNet gait database span roughly 70–99 %, much wider than the differences one would expect between the model families involved. A two-minute recording yields more than a hundred overlapping windows, and if those are divided at random then windows of one subject end up in both training and test, so a classifier can succeed by recognizing the individual instead of the disease, a textbook case of leakage [3] and a failure mode repeatedly documented in medical machine learning [4]. This report therefore asks two questions at once: how well a state-space model classifies these signals, and how much of a reported accuracy is an artifact of the split.

1.1. Related work

Classical pipelines on this database pair hand-crafted statistics of the VGRF channels with Random Forests or kernel SVMs, while deep models (1D-ResNets [5], recurrent networks and transformers) take either the raw signal or sequences of feature vectors. Structured state-space models are a newer alternative: S4 [6] and then Mamba [7] reach transformer-level sequence modeling at linear rather than quadratic cost, because Mamba makes the state transition depend on the input instead of being fixed. It has been applied to speech, EEG and ECG; we are not aware of a VGRF-based PD study that uses it under a subject-disjoint protocol.

1.2. Contributions of this work

We train a Mamba classifier on this dataset and compare it with the Random Forest, SVM-RBF, 1D-ResNet and LSTM baselines from the previous stage of the project, evaluating every model with the same code and the same subject-disjoint folds. Two findings came out of the experiments rather than out of the plan: the input representation mattered more than the architecture, and the split unit mattered more than either. We report the subject-wise numbers as our result and the window-wise ones only as a caution.

2. MATERIALS & METHODS

Figure 1 summarizes the complete pipeline, from the raw force recordings to the per-subject PD/HC decision.

2.1. Data

We use the PhysioNet *Gait in Parkinson's Disease* database [8], pooling its three studies (Ga, Ju, Si). Each subject walked on level ground for about two minutes while eight force sensors per foot sampled at 100 Hz; the dataset provides the 16 sensor channels plus the total force under each foot. Our subset comprises the 100 subjects with a normal-walk recording, balanced by design into 50 PD patients and 50 healthy controls (HC). Table 1 summarizes it: the groups are matched on sex, age and body size and separate on exactly the variables the disease is expected to move, gait speed and the timed up-and-go test. Heights are given in centimeters after harmonizing the two unit conventions of the source file, which records 65 entries in meters and 32 in centimeters.

The course-provided five-fold partition assigns 60 training, 20 validation and 20 test subjects per fold, class-balanced, with test sets disjoint across folds so that pooling them covers every subject exactly once. The folds are not stratified by sub-study, whose PD rates differ (0.58, 0.50 and 0.43 for Ga, Ju and Si), a confound we return to in Section 4.

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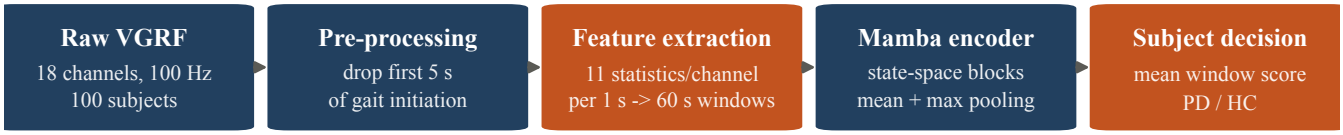


Fig. 1. End-to-end pipeline: raw VGRF recordings become one feature vector per second, grouped into 60 s windows and scored by the Mamba encoder.

Table 1. The 100 subjects: mean ± standard deviation, with p from Welch's t-test (χ^2 for the categorical rows).

	HC (n=50)	PD (n=50)	p
Female / male	25 / 25	25 / 25	1.00
Age (years)	63.0 ± 8.4	66.5 ± 9.8	0.06
Height (cm)	167.6 ± 8.6	166.3 ± 7.9	0.45
Weight (kg)	71.9 ± 12.8	68.5 ± 12.0	0.18
BMI (kg/m^2)	25.5 ± 3.7	24.7 ± 3.1	0.26
Gait speed (m/s)	1.24 ± 0.17	1.02 ± 0.22	<0.001
TUAG (s)	9.2 ± 1.6	12.2 ± 5.0	<0.001
Hoehn & Yahr	—	2.3 ± 0.4	—
UPDRS (total)	—	29.5 ± 11.1	—
Study Ga / Ju / Si	13 / 16 / 21	18 / 16 / 16	0.48

Table 2. The 11 statistics computed per second on each of the 18 channels ($11 \times 18 = 198$ features).

Level	Dispersion & shape	Dynamics & spectrum
mean	standard deviation	zero-crossing rate
median	RMS	spectral energy
minimum	skewness	spectral entropy
maximum	kurtosis	

2.1.1. *Pre-processing*

The first five seconds of each recording are discarded to remove gait initiation. Each window inherits its subject's label, and channel normalization is a per-channel z-score whose mean and standard deviation are computed over the training windows of the current fold only, then applied unchanged to validation and test.

Three input representations were compared, each with its own segmentation. **(i) Raw samples**: the recording is cut into 5 s windows with a 1 s step, giving a 18×500 matrix per window and 10 407 windows in total, between 35 and 254 per subject. **(ii) Amplitude-statistic features**: each second of signal becomes one feature vector holding the 11 statistics of Table 2 for each of the 18 channels, i.e. 198 dimensions; these vectors are then grouped into windows of 30 or 60 seconds with a step of 10 or 15, which yields 827 and 363 windows respectively, between 1 and 14 per subject at 60 s. **(iii) Stride-cycle features**: one feature vector per gait cycle carrying stride, stance and swing time, stance ratio, normalized peak force and impulse, stride-time increment and double-support time for both feet; contacts are detected by thresholding each foot's total force at 5 % of its maximum, and strides outside 0.5–2.5 s are discarded. The reported model uses representation (ii) with 60 s windows.

2.1.2. *Materials/Instruments used*

Models were trained on a Kaggle NVIDIA Tesla T4 (16 GB, compute capability 7.5) with PyTorch 2.10 (CUDA 12.8) and `mamba-ssm` 2.3.2, using mixed precision. Development, feature extraction and evaluation ran locally on a machine with an Apple M1 chip with 8GB RAM. Code was version-controlled on GitHub and pulled by the Kaggle notebook at the start of every run, so each reported number comes from a known commit.

2.2. Proposed approach

2.2.1. *Mamba classifier*

A Mamba block is a selective state-space layer: its state transition and input projections are functions of the input, so the recurrence can forget or retain information conditioned on content, and it runs in linear time in the sequence length [7]. Standard Mamba is causal, which suits language modeling but discards the right-hand context when an entire window is available at inference. We therefore also implemented a bidirectional block (a forward mixer and, after reversal, a second mixer with independent parameters, summed before the residual connection) and ablate it in Section 3.2. The reported model uses it.

The classifier projects each feature vector to d_{model}, applies L such blocks, and pools over time by concatenating the mean and the maximum of the final representation, which a linear layer maps to two logits. Three sizes were studied $(d_{\text{model}}, L) \in \{(64, 2), (128, 4), (256, 6)\}$; the configuration selected in Section 3.3 uses the middle one.

2.2.2. *Training strategy*

Per fold we grid-search dropout $\in \{0.1, 0.3\}$ and learning rate $\in \{3 \cdot 10^{-4}, 10^{-4}\}$, training with AdamW (weight decay 10^{-3}), three warm-up epochs followed by cosine decay, batch size 128 and at most 30 epochs, with early stopping on validation ROC-AUC (patience 6, minimum 8 epochs). The configuration with the highest validation ROC-AUC is retrained on the union of training and validation subjects for the epoch count optimal on validation, and that model predicts the test subjects, the deep-learning equivalent of `GridSearchCV(refit=True)` on a predefined split. Because some folds occasionally converged to a degenerate solution, a fold whose best validation AUC stays below 0.70 is retrained with a different seed (at most three attempts). The final model averages three seeds.

2.2.3. *Decision rule and subject-level aggregation*

The network scores windows, but the clinical unit is the subject, so a subject's score is the mean of its window scores. The decision threshold is calibrated per fold on that fold's *validation* subjects by maximizing balanced accuracy, using the model trained on training data only; test data never influences it. The fixed threshold 0.5 differs by at most two points.

2.2.4. *Deployment on unseen cohorts*

Each fold's weights are exported with its threshold and the per-channel statistics of *that fold's training subjects*. A new recording is normalized with those stored statistics (never with statistics recomputed from the new cohort, which would leak its distribution into its own predictions) and the five folds' scores are averaged.

2.2.5. Evaluation metrics

We report accuracy, precision, recall and F1 (weighted by class support, so recall equals accuracy by construction), with sensitivity, specificity, ROC-AUC and confusion matrices. All values are the mean and standard deviation over 1000 bootstrap resamples that draw whole subjects, so windows of one subject are never counted as independent evidence. Each accuracy is tested against the chance level with a one-sided exact binomial test over the 100 subject decisions, and differences between models, which are evaluated on the same subjects, with DeLong's test for AUC and an exact McNemar test for accuracy.

3. EXPERIMENTS & RESULTS

Every experiment uses the five predefined subject-wise folds of Section 2.1. Roughly twenty GPU sessions covered the five groups of experiments below.

3.1. Experiment 1: what the network is fed

Raw signals were not merely weaker but unstable. On folds 1 and 4 the validation AUC never left the neighborhood of chance (0.53 and 0.54) while folds 2, 3 and 5 exceeded 0.92 with the same code and hyper-parameter grid. Longer windows (30 s, 60 s), restarts with fresh seeds and a larger encoder all failed to repair it. Amplitude-statistic features fixed the problem outright: the weakest fold rose to 0.75 validation AUC. Comparing each representation at its own best configuration, subject-level accuracy on both feet went from 69 % with raw 5 s windows to 80 % with amplitude-statistic features, against 65 % for raw 60 s windows and 66 % for stride-cycle features (best-case figures per representation; the configuration actually reported is chosen in Section 3.3 and scores 75 %). Fusing the stride and amplitude families did not beat the amplitude features alone, even though the two are complementary at subject level in a Random Forest (AUC 0.79, 0.83 and 0.86 for stride, amplitude and both). An unsupervised embedding of the same features averaged per subject (Figure 2) shows the two classes overlapping heavily, with no cluster structure that separates them.

3.2. Experiment 2: architecture

With the input fixed, encoder capacity was the factor that moved accuracy: enlarging the encoder from $d_{\text{model}} = 128$, $L = 4$ to $d_{\text{model}} = 256$, $L = 6$ took subject-level accuracy from 72 % to 80 % on 30 s windows. The bidirectional block did not. Run at matched size, matched input and three seeds each, the causal and bidirectional encoders are indistinguishable: 77 % accuracy and AUC 0.852 for both. We therefore make no claim of a benefit from processing the window in both directions, and note that in the larger encoder size and direction are confounded anyway, since a bidirectional block doubles the mixer parameters.

Training converged very early: the best epoch was typically between one and four out of thirty, a symptom of the small effective sample rather than of the schedule. Stronger regularization (dropout 0.5) did not help, and neither did five seeds instead of three or late fusion across feature families.

3.3. Experiment 3: selecting a configuration without using the test set

Forty-eight configurations were trained in total. Choosing among them by pooled test accuracy would leak the test set into the model definition, even though each individual run keeps train, validation and test apart. We therefore ranked them by mean validation ROC-AUC across the five folds, a quantity that never sees test data, and report the winner of that criterion.

The two criteria agree on the choice that matters (amplitude-statistic features win under both) and differ on the details: validation prefers 60 s windows with the base encoder, pooled test accuracy 30 s windows with the large one. The three accuracy points between them are the optimism a test-based search would have introduced.

Table 3. Subject-level results, subject-wise protocol: mean ± SD over 1000 bootstrap resamples of subjects, p from a one-sided exact binomial test against chance. Best per column in bold.

Model	Data	Acc (%)	Pre (%)	F1	Sen (%)	AUC	p
SVM-RBF	Left	64 ± 5	65 ± 5	0.64 ± 0.05	60 ± 7	0.74 ± 0.05	0.003
	Right	56 ± 5	57 ± 5	0.56 ± 0.05	54 ± 7	0.65 ± 0.05	0.136
	Both	68 ± 5	69 ± 5	0.68 ± 0.05	66 ± 7	0.77 ± 0.05	<0.001
1D-ResNet	Left	68 ± 5	69 ± 5	0.68 ± 0.05	62 ± 7	0.71 ± 0.05	<0.001
	Right	61 ± 5	65 ± 5	0.59 ± 0.05	38 ± 7	0.70 ± 0.05	0.018
	Both	66 ± 5	67 ± 5	0.66 ± 0.05	56 ± 7	0.76 ± 0.05	<0.001
LSTM	Left	70 ± 4	71 ± 4	0.70 ± 0.04	72 ± 6	0.73 ± 0.05	<0.001
	Right	68 ± 5	68 ± 5	0.68 ± 0.05	68 ± 7	0.69 ± 0.05	<0.001
	Both	63 ± 5	64 ± 5	0.63 ± 0.05	54 ± 7	0.67 ± 0.05	0.006
Random Forest	Left	74 ± 4	75 ± 4	0.74 ± 0.04	76 ± 6	0.82 ± 0.04	<0.001
	Right	$\mathbf{78 \pm 4}$	$\mathbf{79 \pm 4}$	$\mathbf{0.78 \pm 0.04}$	74 ± 6	0.80 ± 0.04	<0.001
	Both	73 ± 4	74 ± 4	0.73 ± 0.04	74 ± 6	0.81 ± 0.04	<0.001
Mamba (ours)	Left	74 ± 4	76 ± 4	0.74 ± 0.04	84 ± 5	0.82 ± 0.04	<0.001
	Right	71 ± 4	72 ± 4	0.71 ± 0.04	68 ± 7	0.81 ± 0.04	<0.001
	Both	75 ± 4	77 ± 4	0.75 ± 0.04	$\mathbf{86 \pm 5}$	$\mathbf{0.85 \pm 0.04}$	<0.001

3.4. Experiment 4: comparison with the baselines

Table 3 lists all models under the subject-wise protocol with identical evaluation code and aggregation. On both feet the Mamba model leads every baseline, by two accuracy points and 0.03 AUC over the Random Forest, but the margin is not uniform: the two tie on the left foot and the Random Forest is clearly better on the right (78 % against 71 %).

Two tests keep these numbers in proportion. Since a subject-level decision is one Bernoulli trial per subject, the last column of Table 3 tests each accuracy against the 50 % chance level with a one-sided exact binomial test: every configuration except the SVM-RBF on the right foot ($p = 0.14$) is significantly above chance, so none of the reported results is a coin flip. Comparing models with each other is a paired problem, because all of them are evaluated on the identical 100 subjects; DeLong's test on the ROC curves gives Mamba $+0.088$ AUC over the SVM-RBF ($p = 0.026$), $+0.092$ over the 1D-ResNet ($p = 0.006$) and $+0.181$ over the LSTM ($p < 0.001$), but only $+0.039$ over the Random Forest ($p = 0.15$). Under an exact McNemar test the accuracy gaps are smaller than the sample can resolve, except the $+12$ points over the LSTM ($p = 0.036$). We therefore read the result as Mamba matching the tuned Random Forest and beating the other three, not as a clear win over all baselines.

The error profiles differ in a way that matters more than the averages. Mamba reaches 86 % sensitivity at 64 % specificity, whereas the Random Forest sits near 74 % on both. For a screening tool the asymmetric profile is preferable: of the 50 patients it misses seven, at the cost of 18 healthy subjects flagged for a second look (Figure 4).

3.5. Experiment 5: the evaluation protocol

To measure the influence of the split unit we repeated the evaluation with windows, not subjects, assigned at random to the 60/20/20 splits. Data, features, models, metrics and code were untouched, and the two classical baselines were retrained. Accuracy rises by twenty

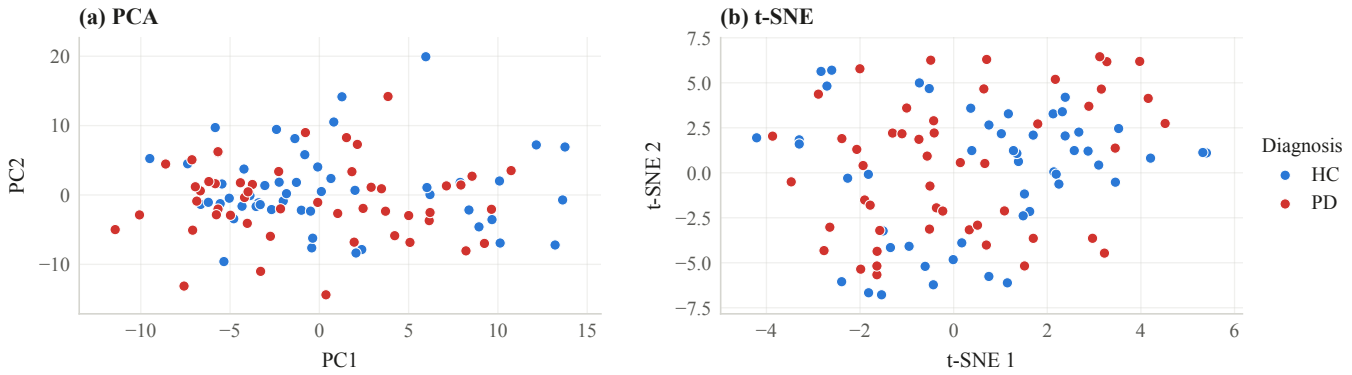


Fig. 2. Unsupervised embedding of the 100 subjects: each point is one subject, described by the 198 amplitude statistics of Table 2 averaged over its recording and z-scored.

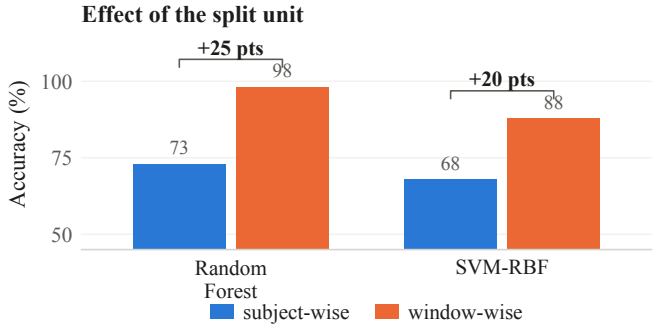


Fig. 3. Effect of the split unit: the same models on both feet under the subject-wise and window-wise protocols, with no other change to features, models or metrics.

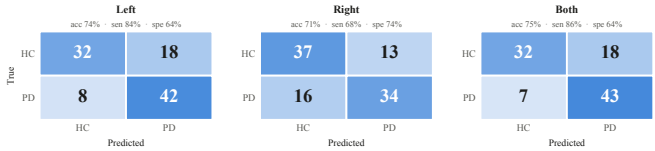


Fig. 4. Subject-level confusion matrices of the reported Mamba model, pooled over the five test folds.

points or more: the Random Forest goes from 73 % to 98 % and the SVM-RBF from 68 % to 88 % (Figure 3). A per-subject decision has no meaning here, since a subject's windows are spread across training and test. These window-level numbers measure how well a model recognizes individuals it has already seen, not whether it detects Parkinson's disease in a new patient; we give them purely as a reference point for the literature, where the SVM-RBF figure of 88 % is close to accuracies frequently reported for this database.

4. DISCUSSION

Mamba and the Random Forest are statistically tied (Section 3.4), the expected outcome on a cohort of this size. Deep sequence models are data-hungry by construction: a Mamba encoder must learn a representation of gait, and the mapping from it to a diagnosis, from 60 training subjects per fold, whereas a Random Forest starts from statistics that already encode what a gait cycle is and has only to weight them. With 100 subjects a well-tuned classical baseline is not a weak opponent but the natural favorite, and a state-space model that matches it is doing so far outside its usual regime.

The two families also differ in what they let a clinician inspect: a Random Forest exposes feature importances, so a decision traces back to a named quantity such as the variability of right-foot stance ratio, whereas the Mamba encoder offers no comparable account of its internal state. Where the two are statistically tied, that transparency is a real argument for the classical model.

The raw-input collapse of Section 3.1 makes the same point. With 60 training subjects a network reading raw traces must discover both what a gait cycle looks like and which of its properties separate patients from controls; the statistics supply the first half, leaving the sequence model to learn how the descriptors evolve. The unstratified sub-studies compound this, since capacity spent on study identity does not transfer to held-out subjects.

A causal scan gives the first time steps of a window almost no context, yet it matched the bidirectional block (Section 3.2). Mean-pooling is the likely explanation: it already aggregates across the window, so the late time steps of a causal scan carry the evidence the classifier needs.

The protocol effect of Section 3.5 has a mechanical explanation: windows cut one second apart are near-duplicates that share sensor placement, body mass and an individual gait signature, so under a random split most test windows have a near neighbor in training and a classifier can identify the person and recall the label.

Several limitations remain. One subject is worth one accuracy point, so the ± 4 spread leaves most per-configuration differences insignificant. Only the normal-walk recordings were used, and stride detection relies on a fixed force threshold, adequate here but in need of revisiting for severely impaired gait.

5. CONCLUSION

We classified Parkinson's disease from vertical ground reaction force recordings with a Mamba state-space model on five subject-disjoint folds. On both feet it reaches 75 ± 4 % accuracy, 0.75 weighted F1 and 0.85 AUC at subject level: ahead of the SVM-RBF, 1D-ResNet and LSTM baselines run through the same evaluation code, and level with a tuned Random Forest. The gain came from the input representation, not the architecture.

The methodological finding may be the more useful one. A window-wise split raises Random Forest accuracy from 73 % to 98 % with no other change to features, models or metrics: when one recording yields more than a hundred correlated windows, an accuracy quoted for this database cannot be interpreted until the split unit is stated. We chose the final configuration on validation AUC for the same reason. Every model plateaued in the same band, pointing at the cohort rather than the method: pooling the dual-task recordings and other public VGRF cohorts is the obvious next step.

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